Light

GCSE Computer Science - Data Representation: Numbers (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What base is the binary number system?

- A. Base 2
- B. Base 8
- C. Base 10
- D. Base 16

Q2 (1 mark)

How many different digits does hexadecimal use?

- A. 2

- B. 8
- C. 10
- D. 16

Q3 (1 mark)

How many bits are in one byte?

- A. 2
- B. 4
- C. 8
- D. 16

Q4 (1 mark)

What is the denary (decimal) value of the binary number 1010?

- A. 5
- B. 20
- C. 12
- D. 10

Q5 (1 mark)

How many bytes are in 1 kilobyte (kB)?

- ○ **A.** 100
- ○ **B.** 512
- ○ **C.** 1,000
- ○ **D.** 1,024

Q6 (1 mark)

Why do computers use binary to represent all data?

- ○ **A.** Binary is easier for humans to read
- ○ **B.** Binary was invented specifically for computers
- ○ **C.** Binary uses less storage than other number systems
- ○ **D.** Electronic circuits can reliably distinguish between two states (on/off)

Q7 (1 mark)

Why is hexadecimal often used by programmers instead of binary?

- ○ **A.** Hexadecimal is faster for computers to process
- ○ **B.** Hexadecimal uses fewer symbols than binary
- ○ **C.** Computers actually store data in hexadecimal

- **D.** It provides a shorter, more readable representation of binary values

Q8 (1 mark)

What denary (decimal) value does the hexadecimal digit 'C' represent?

- **A.** 10
- **B.** 11
- **C.** 12
- **D.** 13

Q9 (1 mark)

What is the maximum denary value that can be represented with 4 bits?

- **A.** 4
- **B.** 8
- **C.** 16
- **D.** 15

Q10 (1 mark)

How many kilobytes are in 1 megabyte (MB)?

- **A.** 100

- B. 500
- C. 1,000
- D. 1,024

Q11 (1 mark)

In an 8-bit binary number, what is the place value of the leftmost bit?

- A. 1
- B. 64
- C. 256
- D. 128

Q12 (1 mark)

What values can a single bit represent?

- A. Any number from 0 to 9
- B. Any number from 0 to 255
- C. Any letter A to Z
- D. Only 0 or 1

Q13 (2 marks)

Calculate the result of adding these two binary numbers: 0110 + 0011

- A. 0111
- B. 1000
- C. 1001
- D. 1010

Q14 (2 marks)

Convert the hexadecimal number 2F to binary.

- A. 0010 1111
- B. 0010 1011
- C. 0011 1111
- D. 0100 1111

Q15 (2 marks)

What is the result of applying a left shift of 2 places to the binary number 00001100?

- A. 00000011
- B. 00011000
- C. 00110000
- D. 01100000

Q16 (2 marks)

Convert the denary number 200 to binary.

- ○ **A.** 1100 0100
- ○ **B.** 1110 0000
- ○ **C.** 1101 0000
- ○ **D.** 1100 1000

Q17 (2 marks)

A file is 2,500 kB. How many MB is this?

- ○ **A.** 0.25 MB
- ○ **B.** 2.5 MB
- ○ **C.** 25 MB
- ○ **D.** 250 MB

Q18 (2 marks)

What is the result of applying a right shift of 3 places to the binary number 01100000?

- ○ **A.** 00001100
- ○ **B.** 00011000
- ○ **C.** 00110000

- **D.** 11000000

Q19 (2 marks)

Convert the denary number 175 to hexadecimal.

- **A.** AB
- **B.** AF
- **C.** BF
- **D.** FA

Q20 (2 marks)

Calculate the result of adding these three binary numbers: 0010 + 0011 + 0100

- **A.** 1000
- **B.** 1010
- **C.** 1001
- **D.** 1011

Q21 (2 marks)

Convert the binary number 1011 0101 to hexadecimal.

- **A.** A5

- B. B5
- C. B6
- D. D5

Q22 (2 marks)

A hard drive has capacity of 2 TB. How many GB is this?

- A. 2,000 GB
- B. 200 GB
- C. 20,000 GB
- D. 2,000,000 GB

Q23 (2 marks)

A programmer wants to multiply a number by 8 quickly. How many left shifts should be applied?

- A. 1
- B. 2
- C. 3
- D. 8

Q24 (3 marks)

Calculate the result of adding these three 8-bit binary numbers: 0011 0010 + 0010 0001 + 0001 1100

- ○ **A.** 0101 0011
- ○ **B.** 0110 1111
- ○ **C.** 0111 0000
- ○ **D.** 1000 0001

Q25 (3 marks)

In web design, colours are often specified using hexadecimal. The colour #3A7F00 represents RGB values. Analyse what each pair of hex digits represents and calculate the green component in denary.

- ○ **A.** Green = 58 (the first pair 3A)
- ○ **B.** Green = 127 (the second pair 7F)
- ○ **C.** Green = 0 (the third pair 00)
- ○ **D.** Green = 163 (calculated from all pairs)

Q26 (3 marks)

A sensor reading gives the binary value 11010000. Analyse what happens when this value undergoes a right shift of 4 places, explaining both the process and the mathematical effect.

- ○ **A.** Result is 00001101 (13); right shift by 4 divides by 16 because 2^4 = 16, so 208/16 = 13
- ○ **B.** Result is 00001101 (13); right shift by 4 divides by 4, so 52/4 = 13

- ○ **C.** Result is 11010000 (208); shifts have no effect on values
- ○ **D.** Result is 00000000 (0); all bits are shifted out

Q27 (3 marks)

A music streaming service stores songs at 320 kB per second of audio. Calculate how many songs averaging 3 minutes each could fit on a 64 GB storage device.

- ○ **A.** Approximately 1,100 songs
- ○ **B.** Approximately 3,500 songs
- ○ **C.** Approximately 11,000 songs
- ○ **D.** Approximately 35,000 songs

Q28 (3 marks)

Calculate the result of adding these three 8-bit binary numbers: 0010 1010 + 0001 0101 + 0011 0011

- ○ **A.** 0110 1100
- ○ **B.** 0111 0000
- ○ **C.** 0111 0010
- ○ **D.** 1000 0010

Q29 (3 marks)

The bit pattern 0100 0001 could represent different things depending on context. Analyse what this pattern represents as: (a) an unsigned

binary number, and (b) an ASCII character.

- ○ **A.** (a) 41, (b) the letter 'a'
- ○ **B.** (a) 65, (b) the letter 'A'
- ○ **C.** (a) 65, (b) the number 1
- ○ **D.** (a) 17, (b) the letter 'A'

Q30 (3 marks)

A memory address is displayed as hexadecimal A3. Convert this to binary and denary, then explain why programmers prefer hexadecimal for memory addresses.

- ○ **A.** Binary: 1010 0011, Denary: 163. Hex is preferred because it is more compact and each hex digit maps directly to 4 binary bits, making conversion simple
- ○ **B.** Binary: 1001 0011, Denary: 147. Hex is preferred because computers process it faster than binary
- ○ **C.** Binary: 1010 0011, Denary: 163. Hex is preferred because it uses fewer symbols than binary
- ○ **D.** Binary: 1011 0010, Denary: 178. Hex is preferred because it is the native format of computer memory

Total: **55** marks • Time: **30 min**

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